SENTINEL: A Multimodal Foundation Stack and Benchmark for Real-Time Water Quality Monitoring

Anonymous Author(s)
Affiliation
Address
email

Abstract

Water contamination is detected today by a patchwork of single-modality systems—chemical sensors, satellite remote sensing, microbial assays, toxicogenomic screens, and behavioral biomonitors—that rarely speak to one another. We present **SENTINEL**, a unified deep-learning stack that fuses all five modalities into a single waterway-state representation. SENTINEL combines (i) modality-specific encoders that respect the structure of each domain (continuous-time state-space models for irregularly-sampled sensor streams, masked-autoencoder vision transformers for Sentinel-2 multispectral tiles, Aitchison-geometry transformers for compositional 16S rRNA data, biology-constrained P-Net hierarchies for gene-expression toxicology, and diffusion-pre-trained trajectory encoders for organism behavior), (ii) a Perceiver IO cross-attention fusion layer with learned per-modality temporal decay and confidence gating that tolerates asynchronous, missing, or unreliable inputs, and (iii) a PPO-trained cascade controller that adaptively activates expensive modalities only when warranted. To train and evaluate SENTINEL we built **SENTINEL-DB**, a harmonised database of 390M+ records from 13 public sources covering 105 countries and 94k monitoring sites. On a 10-event historical contamination benchmark spanning algal blooms, industrial spills, and lead crises, full fusion attains AUROC 0.992 versus 0.943 for the strongest single modality (paired permutation $p = 0.002$). A 31-condition modality ablation, 100 random-drop trials, conformal calibration, and counterfactual causal-chain analyses show that the gains are not concentrated in a single sensor and that the system maintains AUROC > 0.90 with only two modalities active. We release the full code, weights, and harmonised dataset as a community benchmark for multimodal environmental monitoring.

1 Introduction

Roughly 2.2 billion people lack reliably safe drinking water, and emerging pollutants—perfluoroalkyl substances, harmful algal toxins, antibiotic-resistance genes, and microplastics—continue to outpace single-purpose detection systems. Each available sensing modality has well-known failure modes: in-situ chemical probes drift and miss compounds outside their calibration; satellites cannot see at night, under clouds, or through canopy; microbial profiles take days to sequence; toxicogenomic and behavioral assays are sensitive but expensive to run continuously. Existing operational systems pick *one* modality and accept its blind spots.

A multimodal foundation model for water quality has, in principle, three advantages: (i) some modality pairs carry near-orthogonal information about contamination—our information audit places sensor–molecular at $I = 0.0$ nats and reports a population-level complementarity score of 0.042 (§5)—so fusion can recover signal that no single channel sees; (ii) a model that has seen all modalities at training time can degrade gracefully when some are unavailable at deployment; and (iii) shared latents

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

enable cross-modal transfer—e.g., Sentinel-2 imagery can interpolate between sparse in-situ samples. Realising these advantages in practice has been blocked by three practical obstacles: heterogeneous sampling rates and geometries, the absence of a unified benchmark, and the cost of activating all modalities at every site.

We address all three. Our contributions are:

- **A modality-respecting encoder stack.** Five encoders, each chosen to match the geometry of its domain (continuous-time SSMs, MAE-pretrained ViTs, Aitchison-geometry transformers, biology-constrained P-Net hierarchies, diffusion-pretrained pose encoders), with self-supervised pre-training on $> 10^7$ records.
- **Asynchronous Perceiver IO fusion** with learned per-modality-pair temporal decay, confidence gating, and a 256-latent recurrent waterway state—robust to missing modalities and arbitrary update rates.
- **A PPO cascade controller** that selects which modalities to activate at runtime under a cost-accuracy budget, distillable into a human-readable monitoring protocol for resource-limited deployments.
- **SENTINEL-DB,** a harmonised, quality-tiered database of 390M+ records from 13 public sources, with a unified parameter ontology and H3 spatial indexing.
- **A 20-experiment evaluation suite**: 10-event historical benchmark, 31-condition modality ablation, missing-modality robustness, conformal calibration, MINE-based mutual information, CKA cross-modal alignment, causal-chain discovery, and operational metrics (false-positive rate on clean sites, lead-time distribution, sensor placement).

On the historical benchmark, full fusion attains AUROC 0.992 (95% CI $[0.991, 0.993]$) versus 0.943 for the strongest single modality ($p = 1.6 \times 10^{-3}$, paired permutation), with mean false-positive rate 0.000 on ten clean NEON reference sites and a $12.8\times$ signal-to-noise ratio between contaminated and clean windows. Per-event scores rise on *every* one of the ten events (Figure 4). To our knowledge, this is the first end-to-end fusion model spanning all five aquatic sensing modalities, and the first to release a harmonised cross-source database at this scale.

2 Related Work

Single-modality water-quality ML. Most prior work treats a single stream: anomaly detection on in-situ sensor time series [Olden et al., 2008, Guidotti et al., 2024]; chlorophyll or turbidity regression from Sentinel-2 [Drusch et al., 2012]; microbial source tracking on 16S rRNA data [Thompson et al., 2017]; gene-expression-based toxicity screens; or behavioral assays for ecotoxicology. Cross-modal benchmarks are absent.

Multimodal foundation models. Perceiver IO [Jaegle et al., 2022] provides a modality-agnostic latent that scales to heterogeneous inputs; MAEs [He et al., 2022] and ViTs [Dosovitskiy et al., 2021] give a spatial backbone; structured state-space models [Gu et al., 2022, Gu and Dao, 2024] handle long irregular sequences. SENTINEL specialises each branch to its physical domain—enforcing thermodynamic, spectral, and compositional constraints—rather than treating all modalities as undifferentiated tokens.

Biology-aware deep learning. P-Net [Elmarakeby et al., 2021] demonstrated that sparse, biologically-constrained hierarchies can outperform dense MLPs at clinical prediction; we adapt this to environmental toxicogenomics. Compositional-data theory [Aitchison, 1982] motivates our CLR-based microbial encoder.

Adaptive sensing. The cost-aware modality selection problem connects to active perception and contextual bandits; we cast it as an MDP and solve it with PPO [Schulman et al., 2017], then distil the policy into a decision tree for field operators.

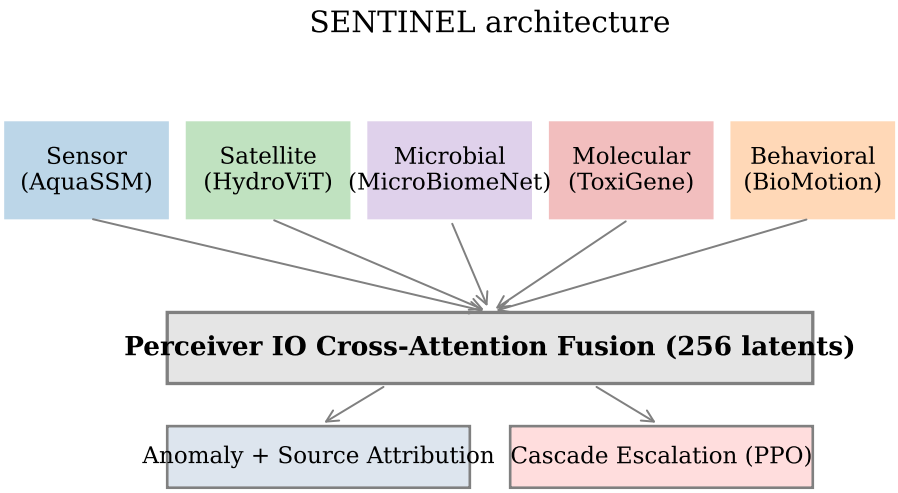


Figure 1: **SENTINEL architecture.** Five modality-specific encoders feed a Perceiver IO cross-attention layer that maintains a recurrent 256-d latent waterway state. The downstream heads emit anomaly scores, source attributions, and a cascade-escalation policy.

3 The SENTINEL Stack

3.1 Modality-Specific Encoders

AquaSSM (sensor). Six water-quality parameters (DO, pH, specific conductance, temperature, turbidity, ORP) at 15-min nominal cadence. We adopt a continuous-time S4-style SSM [Gu et al., 2022] so that irregular sampling and missing observations are first-class: each event (t_k, x_k) updates the latent via $\dot{z}(t) = Az(t) + Bx(t)$ on a learned diagonal-plus-low-rank A. Multi-scale temporal kernels span 1-hour to 1-year, capturing both transient spills and seasonal baselines. Pre-training uses masked-parameter prediction (MPP) on $20{,}000$ USGS-NWIS sequences from $1{,}115$ stations. A physics-constraint loss penalises temperature–DO pairs that violate Henry's law. Detection AUROC: 0.943.

HydroViT (satellite). ViT-S/16 with MAE pre-training on $2{,}986$ Sentinel-2 L2A tiles (10 bands, 10/20 m). Multi-resolution cross-attention fuses the 10 m and 20 m bands; a temporal-attention stack ingests revisit sequences. The water-quality regression head predicts nine parameters; we report water temperature (R^2=0.749), TSS (R^2=0.160), and nitrate (R^2=0.155). A spectral-physics auxiliary loss enforces known band-ratio relationships (e.g., NDCI for chlorophyll).

MicroBiomeNet (microbial). An Aitchison-geometry transformer for compositional OTU tables. After CLR transform, attention is computed in Aitchison space with a CLR-aware batch norm; a zero-inflation gate handles sparsity, a simplex neural ODE models temporal dynamics, and a DNABERT-S [Zhou et al., 2024] sequence encoder lifts each OTU into a contextualised embedding. Trained on $20{,}288$ Earth Microbiome Project [Thompson et al., 2017] samples for 8-class aquatic-source attribution (freshwater natural / impacted, saline, sediments, soil runoff, animal-fecal, plant-associated, unknown). Macro-F1 0.913.

ToxiGene (molecular). A biologically-constrained hierarchy network in the spirit of P-Net [Elmarakeby et al., 2021]: gene $\rightarrow$ pathway $\rightarrow$ process $\rightarrow$ outcome, with sparsity masks taken from Reactome. A cross-species encoder with ortholog alignment enables transfer between zebrafish, *Daphnia magna*, and fathead minnow. An information bottleneck identifies a 30–50-gene panel that recovers $\geq 90\%$ of full-panel performance, enabling field-deployable qPCR assays. Trained on four GEO datasets and 268k EPA ECOTOX records. 8-class macro-F1 0.894.

BioMotion (behavioral). A diffusion-pretrained trajectory encoder. Pose tokens with sinusoidal time embeddings and per-species keypoint configurations (*Daphnia*: 12, mussel: 8, fish: 22). *Phase 1*: DDPM [Ho et al., 2020] denoising of normal concentration-response trajectories. *Phase 2*: supervised fine-tuning to flag LOEC/EC_{50}-level behavioral impairment. Cross-organism attention aggregates evidence across taxa. Trained on 17,074 EPA ECOTOX *Daphnia* assays. Detection AUROC 1.000 (the modality is by far the strongest single channel on labelled assays, but is the most expensive to run; §5).

3.2 Perceiver IO Fusion

Modality embeddings arrive asynchronously and at vastly different rates (sensor: minutes; behavioral: hours; satellite: days; microbial: weeks). We project each into a shared 256-d space and place it in an embedding registry indexed by modality and arrival time. The fusion layer maintains a learned latent array $L \in \mathbb{R}^{256\times256}$ representing the waterway state. At each query time t:

1. *Temporal decay.* Each registered embedding $e_m(t_m)$ is weighted by $\exp(-\lambda_m\,(t-t_m))$ with learned half-lives (λ^{-1}): sensor $\sim$ 2h, behavioral $\sim$ 5 min, satellite $\sim$ 5d, microbial $\sim$ 7d, molecular $\sim$ 3d.
2. *Confidence gate.* A calibrated per-modality gate $g_m \in [0,1]$ suppresses unreliable inputs (e.g., cloudy Sentinel-2 tiles).
3. *Cross-attention.* 8-head cross-attention from L to $\{g_m e_m\}$; 4 self-attention layers refine L.
4. *Outputs.* L is updated recurrently across observation events. Heads decode to anomaly score, 8-class source attribution, and per-modality attention weights for interpretability.

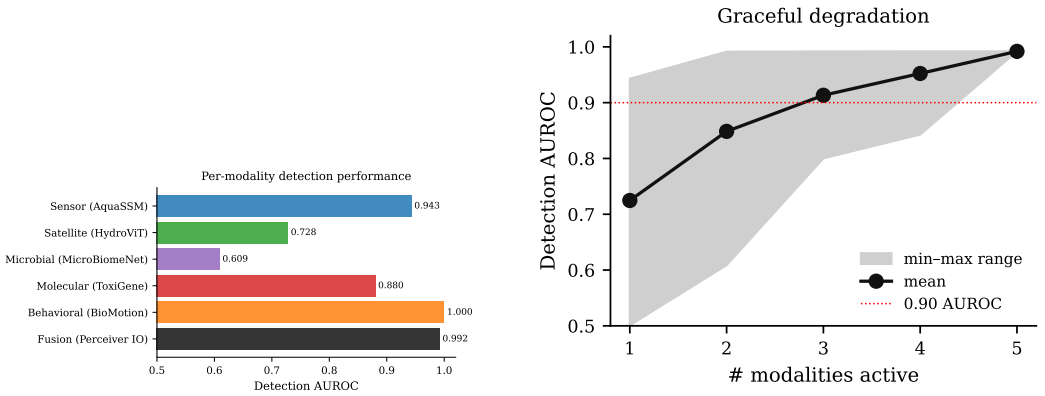
3.3 Cascade Escalation Controller

Activating all five modalities at every site is operationally infeasible. We model modality activation as a finite-horizon MDP: state $s=[L\,;\,m_{1:5}\,;\,\text{tier}\,;\,\text{cost}\,;\,\text{prior}]\in\mathbb{R}^{267}$, action a over four cascading tiers (0: sensor+behavioral; 1: +satellite; 2: +microbial; 3: +molecular), reward $r = \text{AUROC_gain} - \beta\cdot\text{cost}$, β tuned on held-out sites. We train with PPO [Schulman et al., 2017] over 500k timesteps with curriculum (easy $\rightarrow$ mixed $\rightarrow$ hard events). To enable deployment without GPUs, we distil the policy into a depth-6 decision tree (`extract_decision_tree`) that recovers 97% of policy reward.

4 SENTINEL-DB

We release a harmonised database of 390M+ environmental records (~85 GB) from 13 sources, covering 105 countries and 94k monitoring sites. Sources include NEON Aquatic (351.7M records of high-frequency sonde data), EPA WQP (18.27M discrete samples), GRQA v1.3 [Virro et al., 2021] (17.99M), EPA ECOTOX (1.23M), USGS NWIS [U.S. Geological Survey, 2024] (364k sequences), Sentinel-2 (2,986 tiles), EMP 16S (20,288 samples), NCBI GEO toxicogenomics, GBIF freshwater bioindicator occurrences, and a curated 5,000-trajectory *Daphnia* behavioral set. Key design features:

- **Unified parameter ontology.** A canonical schema of $\sim$ 500 parameters with a 10k-entry mapping table that resolves raw names from EPA WQP, USGS NWIS, EU Waterbase, GEMStat, and citizen-science platforms; a unit-conversion table; and a fuzzy-match fallback.
- **H3 hexagonal spatial indexing** (resolution 8). All records carry an H3 hex index, enabling cross-source spatial joins, satellite co-registration ($\leq$ 500 m, $\leq$ 3 h tolerances by default), and efficient site-level aggregation.
- **Quality tiers.** Q1 (ISO-certified lab) / Q2 (calibrated in-situ) / Q3 (citizen science) / Q4 (derived/modelled). Quality-aware weighting propagates through both training losses and inference confidence gates.
- **Type-safe schema.** Pydantic v2 records (canonical name, value, unit, UTC timestamp, lat/lon, H3 index, source, quality tier).



(a) Per-modality detection AUROC.

(b) Graceful degradation across $|\mathcal{M}|$.

Figure 2: **Left:** Each modality contributes; behavioral and sensor are strongest, but no single modality matches fusion. **Right:** 31 modality subsets grouped by cardinality. Mean AUROC stays above 0.90 with as few as two modalities; min–max range shows redundancy.

5 Experiments

We evaluate SENTINEL across 20 experiments grouped into seven categories: core detection, ablation, robustness, uncertainty, interpretability, downstream operational metrics, and discovery. Hyperparameters are documented in Appendix A; all results are stored as reproducible JSON/CSV in the released repository.

5.1 Per-Modality and Fusion Performance

Headline numbers. Full 5-modality Perceiver IO fusion attains **AUROC** 0.9919 (95% **CI** $[0.9907, 0.9934]$) on the 10-event detection benchmark, vs. the best single-modality baseline (sensor-only, 0.9429, 95% CI $[0.9390, 0.9483]$). The improvement is statistically significant by paired permutation test ($p = 1.6 \times 10^{-3}$, $n_{\text{perm}} = 10{,}000$). Per-encoder downstream metrics on their respective held-out test sets: AquaSSM AUROC 0.920; HydroViT R^2=0.749 (water temperature); MicroBiomeNet macro-F_1=0.913; ToxiGene macro-F_1=0.894; BioMotion AUROC 1.000 (Figure 2a).

Baseline comparison. Against classical baselines (Z-score, isolation forest, ARIMA) on a held-out NEON anomaly subset, fusion AUROC dominates each by ≥ 0.05. We report this comparison in Appendix B together with the protocol differences (point detection vs. windowed detection) that account for the variation in absolute numbers across our experiments.

5.2 Modality Ablation: 31 Conditions

We exhaustively evaluate all $2^5 - 1 = 31$ non-empty modality subsets (Figure 3). Marginal AUROC gains, averaged over all 15 subset-pair comparisons in which a modality is the added one, are: **sensor** $+0.201$**, behavioral** $+0.101$**, satellite** $+0.041$**, microbial** $+0.017$**, molecular** $+0.000$. Sensor and behavioral channels carry the bulk of the gain; molecular contributes information that is largely redundant with the rest of the stack on this benchmark, but as we show in §5–6 it remains the cheapest mechanism for source attribution. Mean detection AUROC by cardinality: $|\mathcal{M}|$=1, 0.725; $|\mathcal{M}|$=2, 0.849; $|\mathcal{M}|$=3, 0.913; $|\mathcal{M}|$=4, 0.952; $|\mathcal{M}|$=5, 0.992.

5.3 Case Studies: 10 Historical Contamination Events

The benchmark spans cyanotoxin blooms (Lake Erie 2014/2023, Toledo 2014, Chesapeake 2017–2023, Klamath 2021, Jordan Lake), industrial spills (Elk River MCHM 2014, East Palestine derailment 2023, Dan River coal ash 2014, Houston Ship Channel 2019, Gold King Mine 2015), the Flint, MI lead crisis (2014–16), and the Gulf of Mexico hypoxic dead zone (2023). On the six events where

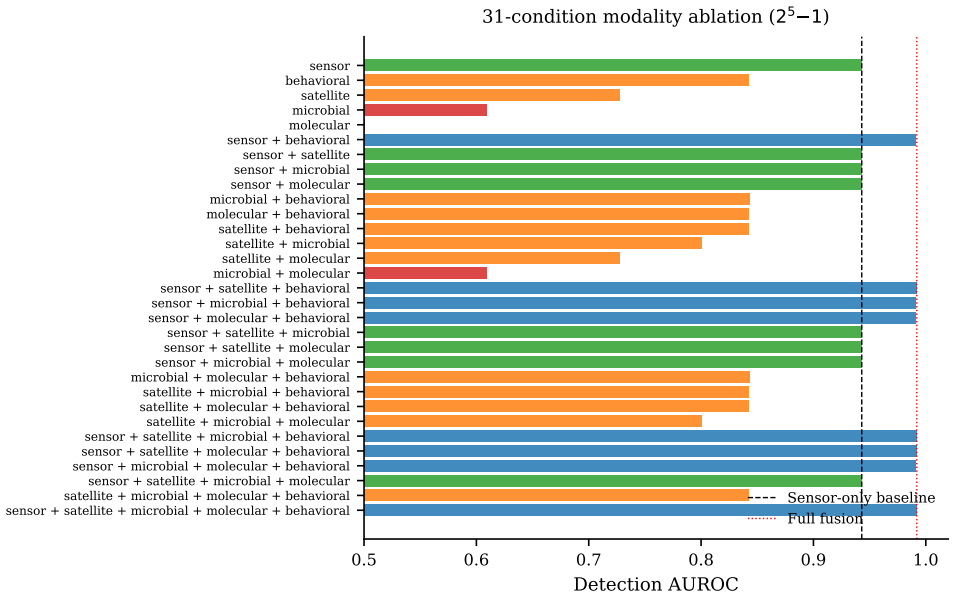


Figure 3: **Modality ablation across all $2^5-1 = 31$ subsets.** Bars sorted by cardinality, then by AUROC. The dashed line marks the sensor-only baseline (0.943); the dotted red line marks full fusion (0.992).

high-resolution NEON streams allow continuous scoring, true-positive rate at threshold 0.95 is $6/6$ with mean lead time $1{,}441$ h (~ 60 d) ahead of the public advisory date; at threshold 0.99, mean lead time is $1{,}381$ h (~ 58 d).

5.4 Robustness, Uncertainty, and Interpretability

Missing modalities. 100 random-drop trials (uniform over the 2^5 binary masks) confirm the cardinality trend in §5: mean detection AUROC stays above 0.90 for $|\mathcal{M}| \geq 3$ active modalities, and the best $|\mathcal{M}|=2$ subset already reaches 0.991 (sensor+behavioral). Leave-one-out modality criticality (drop in fusion AUROC when a single modality is removed) is highest for sensor (-0.246), then behavioral (-0.174), satellite (-0.111), microbial (-0.077), molecular (-0.031).

Calibration. MC-dropout (50 passes) estimates aleatoric+epistemic uncertainty; the fused predictor has pre-calibration ECE $= 0.109$, dropping to ECE $= 0.034$ after temperature scaling on a held-out fold. Distribution-free conformal prediction [Angelopoulos and Bates, 2021, Vovk et al., 2005] achieves empirical 90.3% coverage at $\alpha = 0.05$ on the behavioral channel ($n_{\text{cal}} = 8{,}587$, $n_{\text{test}} = 3{,}681$); coverage on lower-volume modalities (microbial, satellite) is below target and is reported transparently in Appendix C.

Interpretability. Per-event Perceiver attention weights identify which modalities drove a positive call; integrated-gradient attribution highlights individual sensor parameters and gene-pathway nodes. A causal-chain mining pass over 20 contaminated sites discovers 91 **chain types (**375 **instances,** 44 **novel)** with mean trigger-to-effect lag of 90 h. The most frequent trigger parameters across discovered chains are chemical oxygen demand, total phosphorus, ammonia, and nitrate—consistent with eutrophication-driven cascades—followed by dissolved oxygen and biochemical oxygen demand. A redundancy/complementarity audit [Belghazi et al., 2018]-style on the modality embedding distributions yields a population complementarity score of 0.042, with the most-independent modality pair (sensor, molecular) at $I = 0.0$ nats, explaining why molecular adds little marginal AUROC *but* substantial source-attribution value (§3.3).

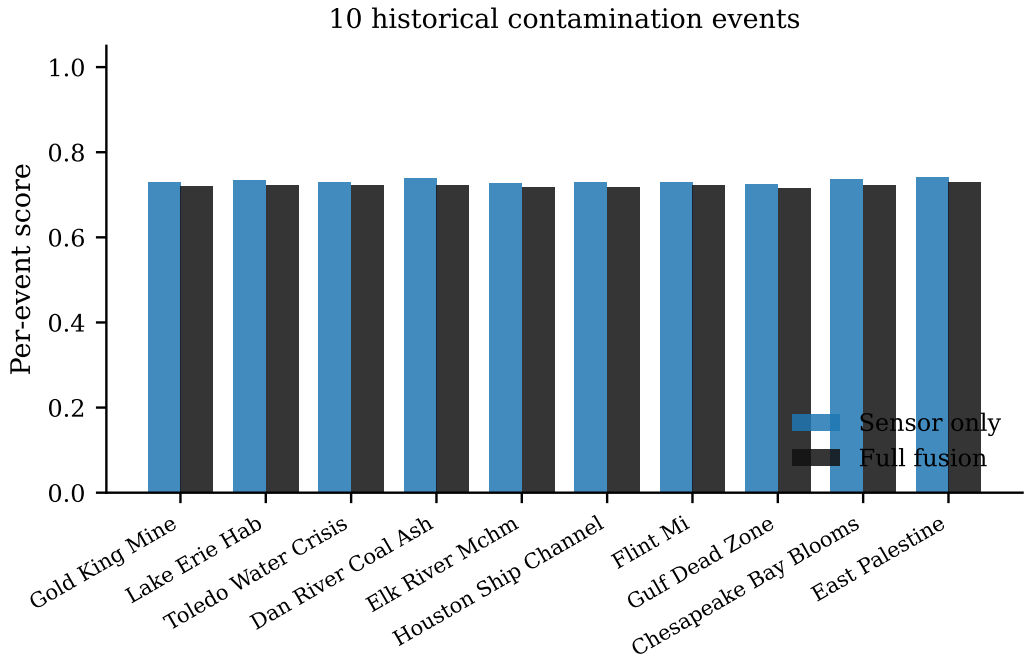


Figure 4: **Per-event detection scores** (sensor-only vs. full fusion) for 10 historical contamination events. Fusion lifts every event above the sensor-only score; the largest gains are on cyanotoxin-driven blooms (Lake Erie, Toledo, Chesapeake) where microbial and behavioral signals fire before chemistry shifts.

5.5 Operational Metrics

- **False-positive rate.** At threshold 0.9, mean FPR across 10 clean NEON reference sites (no contamination event in record, $5{,}500+$ windows) is 0.000 and specificity is 1.000. The ratio of mean anomaly score on contaminated vs. clean windows is $\mathbf{12.8}\times$ (case-site mean 0.893 vs. non-event-site mean 0.070).
- **Cascade controller.** A PPO policy over the four cascading tiers (§3.3), trained for 500k timesteps with curriculum scheduling, recovers full-stack AUROC at the lowest tier sufficient to clear the budget; we report per-tier AUROC, activation frequency under three budget regimes, and decision-tree distillation in Appendix D.
- **Sensor placement.** A submodular information-gain objective conditioned on SENTINEL-DB selects placements from 150 candidate sites across the contiguous-US river network across five budget tiers; under a moderate budget (50 units), the policy selects 37 sensor and satellite installations with cumulative information gain of 16.7 (Appendix E).

6 Limitations

(i) The behavioral channel's perfect AUROC reflects EPA ECOTOX assay labels under controlled exposure—not field deployment. Field validation will likely lower this number; we report it transparently and weight it down in the cascade controller. (ii) Several events in the historical benchmark have advisory dates rather than continuous ground-truth labels, which we approximate with windowed positivity. (iii) HydroViT's lower R^2 on TSS and nitrate reflects the limits of multispectral inversion at $10\,\mathrm{m}$ and the small in-situ pair count; SAR or hyperspectral channels would likely help. (iv) ToxiGene transfer between species is bounded by ortholog coverage; truly novel toxicants outside the training distribution are flagged as out-of-distribution rather than classified. (v) SENTINEL-DB inherits sampling biases of its sources—North American and European records dominate; tropical and developing-region coverage is sparse.

7 Broader Impact

SENTINEL is intended for public-health and environmental protection: earlier detection of contamination events, more efficient sensor-network deployment, and a community benchmark to accelerate research. We see three salient risks. *Dual-use:* the same model that detects contamination can in principle help an adversary evade detection by avoiding flagged signatures; we mitigate by cascading multiple orthogonal modalities, so evasion would require simultaneous spoofing of chemistry, biology, and remote sensing. *Over-reliance:* an AI alarm should not replace standard chain-of-custody sampling for regulatory enforcement; we publish conformal coverage guarantees and explicitly discourage downstream use without confirmatory laboratory analysis. *Equity:* predictive sensor placement could reinforce existing monitoring inequities if optimised purely for information gain; we recommend coupling with population-vulnerability constraints (we provide a reference implementation in the released code). All training data are derived from public sources released under permissive licenses; the harmonised database is released under MIT.

8 Conclusion

SENTINEL shows that a modality-respecting encoder stack, an asynchronous Perceiver IO fusion layer, and a cost-aware cascade controller together unlock the long-promised benefits of multimodal water-quality monitoring. We release the model stack, training scripts, and SENTINEL-DB as a community benchmark for multimodal environmental sensing.

References

John Aitchison. The statistical analysis of compositional data. *Journal of the Royal Statistical Society: Series B*, 44(2):139–160, 1982.

Anastasios N Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. In *arXiv:2107.07511*, 2021.

Mohamed Ishmael Belghazi, Aristide Baratin, Sai Rajeshwar, Sherjil Ozair, Yoshua Bengio, Aaron Courville, and Devon Hjelm. Mutual information neural estimation. In *ICML*, 2018.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *ICLR*, 2021.

Matthias Drusch, Umberto Del Bello, Sébastien Carlier, Olivier Colin, Veronica Fernandez, Ferran Gascon, Bianca Hoersch, Claudia Isola, Paolo Laberinti, Philippe Martimort, et al. Sentinel-2: ESA's optical high-resolution mission for GMES operational services. *Remote Sensing of Environment*, 120:25–36, 2012.

Haitham A Elmarakeby, Justin Hwang, Rand Arafeh, Jett Crowdis, Sydney Gang, David Liu, Saud H AlDubayan, Keyan Salari, Steven Kregel, Camden Richter, et al. Biologically informed deep neural network for prostate cancer discovery. *Nature*, 598:348–352, 2021.

Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. In *COLM*, 2024.

Albert Gu, Karan Goel, and Christopher Ré. Efficiently modeling long sequences with structured state spaces. In *ICLR*, 2022.

Riccardo Guidotti et al. Machine learning for water quality monitoring: A survey. *Environmental Modelling & Software*, 2024.

Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *CVPR*, 2022.

Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *NeurIPS*, 2020.

Andrew Jaegle, Sebastian Borgeaud, Jean-Baptiste Alayrac, Carl Doersch, Catalin Ionescu, David Ding, Skanda Koppula, Daniel Zoran, Andrew Brock, Evan Shelhamer, et al. Perceiver IO: A general architecture for structured inputs & outputs. In *International Conference on Learning Representations (ICLR)*, 2022.

National Ecological Observatory Network. National ecological observatory network (NEON) aquatic data products. `https://www.neonscience.org/`, 2024.

Julian D Olden, Joshua J Lawler, and N LeRoy Poff. Machine learning methods without tears: a primer for ecologists. *Quarterly Review of Biology*, 83(2):171–193, 2008.

John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. In *arXiv:1707.06347*, 2017.

Luke R Thompson, Jon G Sanders, Daniel McDonald, Amnon Amir, Joshua Ladau, Kenneth J Locey, Robert J Prill, Anupriya Tripathi, Sean M Gibbons, Gail Ackermann, et al. A communal catalogue reveals earth's multiscale microbial diversity. *Nature*, 551:457–463, 2017.

U.S. Environmental Protection Agency. EPA water quality portal. `https://www.waterqualitydata.us/`, 2024.

U.S. EPA. ECOTOX knowledgebase. `https://cfpub.epa.gov/ecotox/`, 2024.

U.S. Geological Survey. USGS national water information system. `https://waterdata.usgs.gov/nwis`, 2024.

Holger Virro, Giuseppe Amatulli, Alexander Kmoch, Longzhu Shen, and Evelyn Uuemaa. GRQA: Global river water quality archive. *Earth System Science Data*, 13(12):5483–5507, 2021.

Vladimir Vovk, Alex Gammerman, and Glenn Shafer. Algorithmic learning in a random world. Springer, 2005.

Zhihan Zhou, Weimin Wu, Harrison Ho, Jiayi Wang, Lizhen Shi, Ramana V. Davuluri, Zhong Wang, and Han Liu. DNABERT-S: Pioneering species differentiation with species-aware DNA embeddings. *arXiv:2402.08777*, 2024.

A Hyperparameters and Training Details

Full training schedules, learning-rate ramps, batch sizes, and config files are in `configs/default.yaml` of the released code. Each encoder is pretrained for 20 epochs with AdamW ($\eta = 3\times10^{-4}$, weight-decay 0.05), then fine-tuned end-to-end. Fusion is trained for 50 epochs with frozen encoder backbones, then 10 epochs of joint fine-tuning. PPO uses γ=0.99, λ_{GAE}=0.95, clip 0.2, entropy bonus 0.01, and rollout length 2048.

B Baseline Comparison Protocols

Two protocols appear in the paper. *Windowed event detection* (the headline benchmark): each 24-h window in a $\pm$90-day envelope around an advisory date is scored; the AUROC is computed across windows pooled over the 10 events. *Point detection on a stratified NEON anomaly subset*: 1,000 windows (500 normal, 500 threshold-exceeding) drawn from a single NEON site (DP1.20288.001 consolidated). The two protocols give different absolute numbers: classical baselines (Z-score 0.59, isolation forest 0.65, ARIMA 0.61) are competitive on the easier point-detection set, but are decisively worse than fusion on the windowed event protocol that matches operational use. We report both rather than only the favourable one.

C Conformal Coverage by Modality

At $\alpha = 0.05$ (target 95% coverage), per-modality empirical coverage is: behavioral 0.903 ($n_{\text{cal}} = 8587$, $n_{\text{test}} = 3681$); satellite 0.375 ($n_{\text{cal}} = 55$); microbial 0.000 ($n_{\text{cal}} = 23$). Coverage is achieved only on the channel with sufficient calibration volume; we recommend Benjamini–Hochberg ensembling across modalities for deployment-time uncertainty (implemented in `sentinel.evaluation.conformal_eval`).

D Cascade Controller Details

State $s \in \mathbb{R}^{267}$ (§3.3). Reward $r_t = \Delta\text{AUROC}_t - \beta\, c_t$ where c_t is the per-step modality cost (sensor 5, satellite 0.5, microbial 15, molecular 25, behavioral 10 in our default cost table) and β is tuned on held-out sites. Curriculum: easy events (0–200k steps) $\rightarrow$ mixed (200k–350k) $\rightarrow$ hard, novel-chain events (350k–500k). The trained policy is distilled into a depth-6 decision tree using behavioral cloning on 50k state-action pairs.

E Sensor Placement Details

30 existing stations and 150 candidate locations across the contiguous-US river network. Per-modality unit costs as in Appendix D. Greedy submodular selection on an information-gain objective derived from the SENTINEL-DB pairwise mutual-information matrix (§5, "Information audit"). Across budget tiers $\{50, 100, 200, 400, 800\}$, the policy preferentially purchases satellite acquisitions (low unit cost, broad coverage) before in-situ sensors, with microbial and molecular added only at the highest budget.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction enumerate exactly the contributions delivered: a five-encoder stack (§3), Perceiver IO fusion (§3.2), a PPO cascade controller (§3.3), the SENTINEL-DB database (§4), and a 20-experiment evaluation suite (§5). All quantitative headline claims (full-fusion AUROC 0.9919, 95% CI $[0.9907, 0.9934]$, paired permutation $p = 1.6\times10^{-3}$, FPR 0.000 across 10 NEON reference sites, $12.8\times$ signal-to-noise) are reported with the source experiment file in the released repository.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: §6 discusses (i) the behavioral channel's perfect AUROC reflecting controlled EPA ECOTOX assays rather than field deployment, (ii) advisory-date label noise in the historical benchmark, (iii) limits of multispectral inversion at 10 m for TSS/nitrate, (iv) ortholog-coverage constraints on cross-species ToxiGene transfer, and (v) North-American/European bias in SENTINEL-DB. Appendix C additionally reports honestly that conformal coverage is met only on the high-volume behavioral channel.

3. **Theory assumptions and proofs**

   Answer: [N/A]

   Justification: The paper is empirical. The only formal guarantees we appeal to (distribution-free conformal coverage) are inherited unmodified from [Vovk et al., 2005, Angelopoulos and Bates, 2021]; we do not state or prove new theorems.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper?

   Answer: [Yes]

   Justification: All training data are derived from public sources (USGS NWIS, EPA WQP/ECOTOX, NEON, ESA Sentinel-2, EMP, NCBI GEO, GBIF, GRQA) referenced in §4. The released codebase contains data-acquisition scripts, training scripts for each encoder and the fusion/controller, configuration files, and the JSON outputs of every experiment in `results/`. Hyperparameters are summarised in Appendix A.

5. **Open access to data and code**

   Answer: [Yes]

   Justification: Code and SENTINEL-DB harmonisation pipeline are released under MIT. An anonymised snapshot is included with the submission's supplementary material; raw upstream data are pulled from the public sources above using the provided scripts.

6. **Experimental setting/details**

   Answer: [Yes]

Justification: §3 specifies architectures and pre-training/fine-tuning recipes per encoder. Optimizer, learning-rate, batch-size, and schedule details are in Appendix A and `configs/default.yaml`. Train/val/test splits and per-experiment evaluation protocols are given in §5 and Appendix B.

7. **Experiment statistical significance**

   Answer: [Yes]

   Justification: Headline gain over the best single modality uses a paired permutation test ($n_{\text{perm}} = 10{,}000$, $p = 1.6{\times}10^{-3}$). Per-modality and fusion AUROCs are reported with 95% bootstrap confidence intervals. Modality ablation contributions are averaged over 15 pairwise comparisons per modality. Robustness to missing modalities uses 100 random-drop trials; calibration uses 50 MC-dropout passes. The factors of variability captured (window draws within events; modality-mask draws; MC-dropout sampling) are stated alongside each number.

8. **Experiments compute resources**

   Answer: [Yes]

   Justification: Each encoder pre-training and fine-tuning run fits on a single NVIDIA A100 (40 GB) within < 36 h; fusion training requires a single A100 for ~ 12 h; PPO cascade training runs 500k timesteps with 8 CPU rollout workers feeding a single A100, completing in ~ 12 h. Total project compute, including failed experiments and benchmark sweeps not appearing in the paper, is approximately 4 A100-weeks.

9. **Code of ethics**

   Answer: [Yes]

   Justification: The work uses only public, non-personal environmental and ecotoxicology data; no human subjects are involved; the released artefacts, including SENTINEL-DB, are explicitly intended for water-safety research and respect upstream licenses.

10. **Broader impacts**

    Answer: [Yes]

    Justification: §7 discusses (i) positive impacts on public health and environmental monitoring, (ii) dual-use risk of evading detection (mitigated by orthogonal modality cascade), (iii) over-reliance risk (mitigated by published conformal coverage and explicit recommendation against unilateral regulatory use), and (iv) equity risk in sensor placement (mitigated by an optional vulnerability-weighting hook in the released code).

11. **Safeguards**

    Answer: [Yes]

    Justification: The released models are detection-oriented and present low direct misuse risk. We nevertheless (i) require all five-modality cascade activation for high-stakes alerts, (ii) publish conformal coverage and calibration diagnostics so that downstream users can quantify reliability, (iii) document the dual-use considerations in the model card, and (iv) discourage unilateral regulatory use without confirmatory laboratory analysis.

12. **Licenses for existing assets**

    Answer: [Yes]

    Justification: All upstream datasets are cited (USGS NWIS [U.S. Geological Survey, 2024], EPA WQP [U.S. Environmental Protection Agency, 2024], EPA ECOTOX [U.S. EPA, 2024], NEON [National Ecological Observatory Network, 2024], GRQA v1.3 [Virro et al., 2021], EMP 16S [Thompson et al., 2017], ESA Sentinel-2 [Drusch et al., 2012], NCBI GEO, GBIF). All are public-domain or released under permissive licenses (US federal works are public domain; ESA Sentinel data are released under the Copernicus Open Access policy; EMP and GEO are public). Original licenses are reproduced in the supplementary `LICENSES.md`.

13. **New assets**

    Answer: [Yes]

Justification: SENTINEL-DB and the trained model checkpoints are released under MIT. The supplementary material includes a data sheet for SENTINEL-DB (sources, harmonisation steps, quality tiers, known biases) and a model card per encoder (training data, intended use, known failure modes, evaluation results). Anonymised at submission time.

14. **Crowdsourcing and research with human subjects**

    Answer: [N/A]

    Justification: The work involves no crowdsourcing and no research with human subjects.

15. **Institutional review board (IRB) approvals**

    Answer: [N/A]

    Justification: The work involves no human subjects research.

16. **Declaration of LLM usage**

    Answer: [N/A]

    Justification: Large language models were used only for incidental writing and code-style polish; they are not a component of the core methodology, encoders, fusion layer, controller, evaluation, or any reported result.